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Machine learning-based in-hospital mortality prediction models for patients with acute coronary syndrome

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A B S T R A C T

Objectives: The purpose of this study is to identify the risk factors of in-hospital mortality in patients with acute coronary syndrome (ACS) and to evaluate the performance of traditional regression and machine learning prediction models.

Methods: The data of ACS patients who entered the emergency department of Fujian Provincial Hospital from January 1, 2017 to March 31, 2020 for chest pain were retrospectively collected. The study used univariate and multivariate logistic regression analysis to identify risk factors for in-hospital mortality of ACS patients. The traditional regression and machine learning algorithms were used to develop predictive models, and the sensitivity, specificity, and receiver operating characteristic curve were used to evaluate the performance of each model.

Results: A total of 6482 ACS patients were included in the study, and the in-hospital mortality rate was 1.88%. Multivariate logistic regression analysis found that age, NSTEMI, Killip III, Killip IV, and levels of D-dimer, cardiac troponin I, CK, N-terminal pro-B-type natriuretic peptide (NT-proBNP), high-density lipoprotein (HDL) cholesterol, and Stains were independent predictors of in-hospital mortality. The study found that the area under the receiver operating characteristic curve of the models developed by logistic regression, gradient boosting decision tree (GBDT), random forest, and support vector machine (SVM) for predicting the risk of in-hospital mortality were 0.884, 0.918, 0.913, and 0.896, respectively. Feature importance evaluation found that NT-proBNP, D-dimer, and Killip were top three variables that contribute the most to the prediction performance of the GBDT model and random forest model.

Conclusions: The predictive model developed using logistic regression, GBDT, random forest, and SVM algorithms can be used to predict the risk of in-hospital death of ACS patients. Based on our findings, we recommend that clinicians focus on monitoring the changes of NT-proBNP, D-dimer, Killip, cTnI, and LDH as this may improve the clinical outcomes of ACS patients.

1. Background

Acute coronary syndrome (ACS) is the unstable and progressive stage of coronary heart disease, defined as ST-segment elevation myocardial infarction (STEMI), non-STEMI (NSTEMI), and unstable angina (UA) [1-3]. The most common symptom of ACS is chest pain, which is also one of the most common reasons for visiting the emergency room, accounting for 6% of the visits and 27% of medical admissions [4]. Even if ACS patients receive timely percutaneous coronary intervention and/or appropriate antiplatelet drugs, their risk of mortality and disease progression are still high [2,5]. ACS is associated with high

Abbreviations: ACS, acute coronary syndrome; STEMI, ST-segment elevation myocardial infarction; NSTEMI, non-STEMI; UA, unstable angina; LR, logistic regression; GBDT, gradient boosting decision tree; RF, random forest; SVM, support vector machine; cTnI, cardiac troponin I; CK, creatine kinase; NT-proBNP, N-terminal pro-B-type natriuretic peptide; LDH, lactate dehydrogenase; HDL cholesterol, high-density lipoprotein cholesterol; LDL cholesterol, low-density lipoprotein cholesterol; ACEI, angiotensin converting enzyme inhibitor; PCI, percutaneous coronary intervention.

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morbidity and mortality and causes approximately 7 million deaths annually in the Asia-Pacific region [6,7]. Many ACS patients die during hospitalization [8,9]. Even with medical intervention, the mortality risk within 1 year after the diagnosis of ACS remains about 5% [10-12]. Because appropriate management can significantly improve the prognosis of the patients, it is very important to identify ACS patients and their risk of disease progression promptly and accurately. Predicting the mortality risk of ACS patients may help develop appropriate treatment strategies for patients [5,13].

The Global Registry of Acute Coronary Events (GRACE) and Thrombolysis In Myocardial Infarction risk scores are the most popular mortality risk prediction tools of ACS [14-16]. Although these risk assessment tools have been developed and validated, their use in daily practice is still limited. Other cardiovascular disease (CVD) risk prediction models were developed based on machine learning, such as random forest (RF), neural network, and support vector machine (SVM) to solve the limitations of traditional regression-based CVD prediction models [17-19]. However, there are few studies on machine learning models for predicting mortality risk of ACS patients. The purpose of this study is to build predictive models of in-hospital mortality risk of ACS patients based on traditional regression and machine learning algorithms and to evaluate the predictive efficacy of the models.

2. Methods

2.1. Patient and study design

Patients who were admitted to the Chest Pain Center (Emergency Department), Fujian Provincial Hospital from January 1, 2017 to March 31, 2020 due to chest pain were included in the retrospective cohort study. The inclusion criteria included age ≥ 18 years, admission to the hospital with a diagnosis of ACS (STEMI, NSTEMI, or UA), and complete outcome information (cure, improvement, or death) of the patient at the time of the last discharge. Exclusion criteria included pregnant women; patients with malignant tumors, end-stage renal disease, severe liver disease, or blood disease; and patients with non-cardiogenic chest pain. The study was approved by the Ethical Committee of Fujian Provincial Hospital. The informed consent requirement was waived since the study only involved the use of past clinical data.

2.2. Definition and variables

In-hospital mortality was defined as all-cause death during hospitalization. The variables collected include demographic characteristics, comorbidities, thrombolytic therapy, laboratory test data, and physical examination data. They were obtained from the patients' electronic medical records, hospital information system, laboratory information management system, and clinical data repository.

Disease diagnosis adopts the tenth revised edition of the International Classification of Diseases (ICD), referred to as ICD-10. The diagnostic code for acute coronary syndrome is I24.901.

2.3. Construction of prediction model of in-hospital mortality risk for ACS patients

To develop the mortality prediction model of ACS patients, we employed four machine learning algorithms, including logistic regression (LR), gradient boosting decision tree (GBDT), random forest (RF), and SVM. As a linear model, LR was employed as the baseline model. SVM is a classical machine learning algorithm and makes prediction by means of learning the boundary hyperplane in the feature space between positive and negative samples. GBDT and RF are both ensemble methods based on the decision tree model, a flow chart with rules learned from data. In brief, GBDT builds an ensemble of decision trees with each tree trying to fit residual errors of all previous trees in a gradient descent manner, while RF builds an ensemble of decision trees in parallel and hence reduces the prediction error of a single decision tree.

First, the researchers divided the data into training and testing sets according to 7:3. The training data set was used for statistical analysis, feature selectin and model training, while the independent testing data set was used to evaluate the trained models. Second, we performed univariate analysis and chose the most eligible variables for model development. The preprocessing procedure also included mean value imputation, scaling and one-hot encoding. Third, in order to reduce overfitting and improve model accuracy, we split the training data set in a cross-validation scheme and tuned the hyperparameters of each machine learning model to optimize the cross-validation performance. The optimal hyperparameters were then used to fit all training data to obtain the final models. Finally, we calculated a series of metrics to evaluate model evaluation using the independent test set, including the areas under the receiver operating characteristic (AUROC) curves, sensitivity, and specificity.

All machine learning models were developed using Python language (3.7). LR, RF and SVM were implemented using Python library scikit-learn (0.23.2), while GBDT was implemented using Python library Xgboost (1.2.1).

2.4. Statistical analysis

Continuous variables were summarized as median and interquartile range, and categorical variables were expressed by frequency and percentage. Comparisons between training and testing sets for continuous variables and categorical variables were conducted by Mann-Whitney *U* test and Chi-squared test or Fisher's exact test, respectively. Logistic regression analysis was used to analyze the potential risk factors of in-hospital mortality of ACS patient. The variables with $P < 0.1$ in univariable logistic model were inputted in multivariable logistic model and forward selection method was used to identify the factors associated with in-hospital mortality. To assess the predictive performance for in-hospital mortality, logistic regression, random forest, SVM, and GBDT models were constructed using the training set and then evaluated with the testing set. ROC curves were plotted, and the area under the curve (AUC, 95% CI) with associated *P* values was calculated. In addition, sensitivity and specificity for various models were also derived by maximizing the Youden index. Furthermore, top 10 features of importance from the GBDT and random forest models were determined.

3. Results

3.1. Patients and outcomes

A total of 6482 patients were included in the study, including 815 STEMI patients, 1588 NSTEMI patients, and 4079 UA patients. The mortality rate was 1.88% (122/6482). A total of 4537 patients (85 death and 4452 survivors) were assigned into the training set and 1945 cases into the testing set (37 deceased and 1908 survivors). The flow chart of our study was shown in Fig. 1. Clinical characteristics of patients in the mortality and non-mortality groups in the training and testing sets were showed in *Supplemental Table S1*.

3.2. Comparison of baseline characteristics of the patients in the training and testing sets

We compared the baseline characteristics of the patients in the training and testing sets. The results showed that there were no significant differences in the baseline characteristics of patients in the training and testing sets (Table 1).

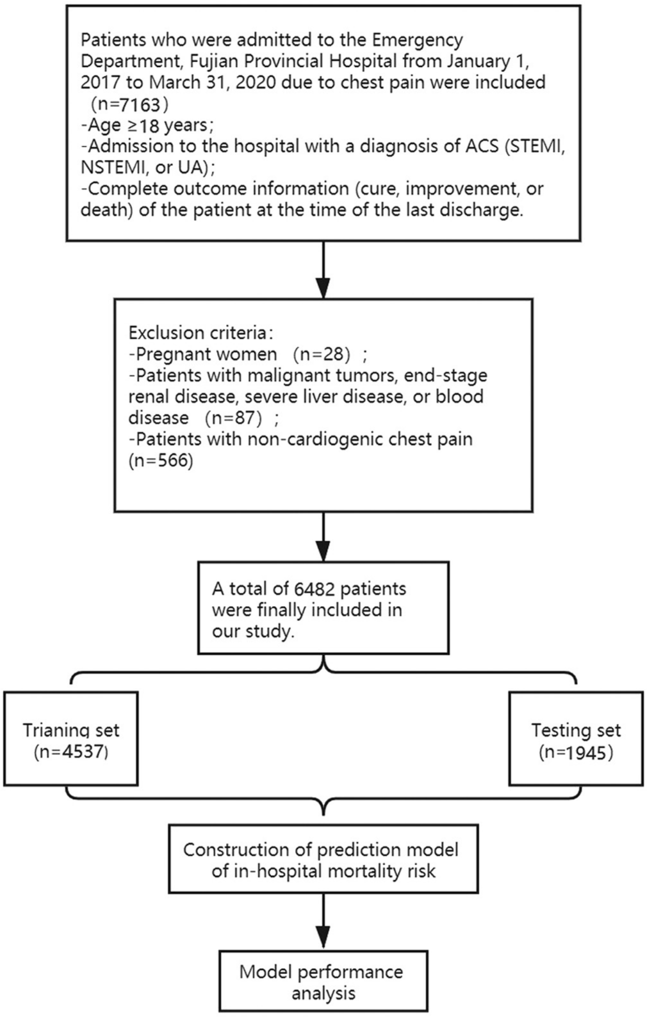

Fig. 1. Flow chart in our study.

Table 1
Patient characteristics in training and validation sets.

		Training (n = 4537)	Testing (n = 1945)	P value
Age	Median (IQR)	67(60, 75)	67(59, 75)	0.574
Gender	Female	1150 (25.3%)	467 (24.0%)	0.254
	Male	3387 (74.7%)	1478 (76.0%)	
Diagnosis	UA	2853 (62.9%)	1226 (63.0%)	0.076
	NSTEMi	1137 (25.1%)	451 (23.2%)	
	STEMI	547 (12.1%)	268 (13.8%)	
Killip	I	1506 (33.2%)	699 (35.9%)	0.186
	II	2357 (52.0%)	973 (50.0%)	
	III	457 (10.1%)	181 (9.3%)	
	IV	217 (4.8%)	92 (4.7%)	
D-Dimer	Median (IQR)	0.33 (0.18, 0.73)	0.33 (0.19, 0.71)	0.793
cTnI	Median (IQR)	0.07 (0.01, 0.45)	0.09 (0.01, 0.57)	0.035
CK	Median (IQR)	83 (58, 134)	84 (59,136)	0.387
NT-proBNP	Median (IQR)	247.3 (74.65,867.7)	245.1 (78.69, 947.0)	0.458
LDH	Median (IQR)	180 (155, 231)	181 (154, 236)	0.823
HDL cholesterol	Median (IQR)	0.98 (0.82, 1.18)	0.99 (0.83, 1.19)	0.312
LDL cholesterol	Median (IQR)	2.36 (1.79, 3.09)	2.37 (1.79, 3.08)	0.945
Total cholesterol	Median (IQR)	3.81 (3.17, 4.62)	3.82 (3.16, 4.63)	0.891
Triglycerides	Median (IQR)	1.34 (1.00, 1.91)	1.34 (0.99, 1.88)	0.513
LVEF	Median (IQR)	59.00 (56.00, 61.00)	59 (56.00, 61.00)	0.459
Hypertension	No	1329 (29.3%)	594 (30.5%)	0.314
	Yes	3208 (70.7%)	1351 (69.5%)	
Diabetes	No	2914 (64.2%)	1277 (65.7%)	0.27
	Yes	1623 (35.8%)	668 (34.3%)	
Clopidogrel	No	611 (13.5%)	279 (14.3%)	0.347
	Yes	3926 (86.5%)	1666 (85.7%)	
Beta-blockers	No	1115 (24.6%)	472 (24.3%)	0.791
	Yes	3422 (75.4%)	1473 (75.7%)	
Calcium channel blockers	No	2906 (64.1%)	1267 (65.1%)	0.401
	Yes	1631 (35.9%)	678 (34.9%)	
Statins	No	132 (2.9%)	50 (2.6%)	0.449
	Yes	4405 (97.1%)	1895 (97.4%)	
Low molecular weight heparin	No	1764 (38.9%)	753 (38.7%)	0.9
	Yes	2773 (61.1%)	1192 (61.3%)	
Ticagrelor	No	3540 (78.0%)	1499 (77.1%)	0.397
	Yes	997 (22.0%)	446 (22.9%)	
Glycoprotein IIb/IIIa receptor antagonist	No	3875 (85.4)	1623 (83.4)	0.043
	Yes	662 (14.6)	322 (16.6%)	
Angiotensin receptor antagonist	No	3155 (69.5)	1344 (69.1%)	0.725
	Yes	1382 (30.5)	601 (30.9%)	
ACEI	No	2077 (45.8)	886 (45.6%)	0.867
	Yes	2460 (54.2)	1059 (54.4%)	
Aspirin	No	458 (10.1)	176 (9.0%)	0.194
	Yes	4079 (89.9)	1769 (91.0%)	
PCI	No	4056 (89.4)	1764 (90.7%)	0.114
	Yes	481 (10.6)	181 (9.3%)	
Thrombolysis with drugs	No	4535 (100.0%)	1943 (99.9%)	0.383
	Yes	2 (0.0%)	2 (0.1%)	
Mortality	No	4452(98.1%)	1908(98.1%)	0.938
	Yes	85(1.9%)	37(1.9%)	

Note: cTnI: cardiac troponin I; CK: creatine kinase; NT-proBNP: N-terminal pro-B-type natriuretic peptide; LDH: lactate dehydrogenase; HDL cholesterol: high-density lipoprotein cholesterol; LDL cholesterol: low-density lipoprotein cholesterol; LVEF: left ventricular ejection fraction; ACEI: angiotensin converting enzyme inhibitor; PCI: percutaneous coronary intervention.

3.3. Univariate and multivariate logistic analysis of risk factors

Univariate analysis found that age, NSTEM, STEMI, Killip III, Killip IV, D-dimer, cardiac troponin I (cTnI), creatine kinase (CK), N-terminal pro-B-type natriuretic peptide (NT-proBNP), lactate dehydrogenase (LDH), high-density lipoprotein (HDL) cholesterol, low-density lipoprotein (LDL) cholesterol, total cholesterol, triglycerides, LVEF, clopidogrel, beta-blockers, calcium channel blockers, statins, ticagrelor, angiotensin converting enzyme inhibitor, and aspirin were significantly associated with in-hospital mortality of ACS patients (Table 2).

After adjusting for confounding factors, the study found that age (OR = 1.042, 95%CI: 1.016,1.068; P = 0.001), NSTEMI (OR = 2.818, 95%CI: 1.505, 5.277; P = 0.01), Killip III (OR = 4.639, 95%CI: 1.839, 11.689; P = 0.001), Killip IV (OR = 8.519, 95%CI: 3.357, 21.618; $P < 0.001$), D-Dimer (OR = 1.098, 95%CI: 1.017, 1.185; P = 0.017), cTnI (OR = 1.085, 95%CI: 1.046,1.125; $P < 0.001$), CK (OR = 1.003, 95%CI: 1.001, 1.006; P = 0.005), and NT-proBNP (OR = 1.006, 95%CI: 1.003,1.008; $P < 0.001$), were independently associated with increased risk of in-hospital mortality (Table 2). Moreover, HDL cholesterol (OR = 0.895, 95%CI: 0.826, 0.969; P = 0.006), and Statins (OR = 0.415, 95%CI: 0.182, 0.948; P = 0.037) were independently associated with decreased risk of in-hospital mortality (Table 2).

Table 2
Univariate and multivariate logistic analysis of risk factors for in-hospital mortality of ACS patients.

	Unadjusted			Adjusted		
Variables	OR	95% CI	P value	OR	95% CI	P value
Age (Unit = 1)	1.108	(1.082,1.135)	<0.001	1.042	(1.016,1.068)	0.001
Gender (Male vs Female)	1.183	(0.707,1.979)	0.522			
Diagnosis:UA vs NSTEMi	7.293	(4.298,12.374)	<0.001	2.818	(1.505,5.277)	0.01
Diagnosis:UA vs STEMI	3.631	(1.783,7.397)	<0.001	1.987	(0.839,4.708)	0.119
Killip: II VS I	2.11	(0.903,4.930)	0.085	1.835	(0.75,4.487)	0.18
Killip: III VS I	10.83	(4.596,25.522)	<0.001	4.639	(1.839,11.698)	0.001
Killip: IV VS I	38.406	(16.750,88.063)	<0.001	8.519	(3.357,21.618)	<0.001
D-Dimer (Unit = 1)	1.244	(1.172,1.321)	<0.001	1.098	(1.017,1.185)	0.017
cTnI (Unit = 1)	1.046	(1.025,1.067)	<0.001	1.085	(1.046,1.125)	<0.001
CK (Unit = 10)	1.004	(1.002,1.005)	<0.001	1.003	(1.001,1.006)	0.005
NT-proBNP (Unit = 100)	1.013	(1.011,1.014)	<0.001	1.006	(1.003,1.008)	<0.001
LDH (Unit = 10)	1.008	(1.005,1.011)	<0.001	1.008	(1.013,1.025)	<0.001
HDL cholesterol (Unit = 0.1)	0.041	(0.018,0.097)	<0.001	0.041	(0.826,0.969)	0.006
LDL cholesterol (Unit = 1)	0.535	(0.409,0.701)	<0.001	0.535		
Total cholesterol (Unit = 1)	0.597	(0.474,0.752)	<0.001	0.597		
Triglycerides (Unit = 1)	1.138	(1.015,1.275)	0.027	1.138		
LVEF (Unit = 1)	0.93	(0.911,0.949)	<0.001			
Hypertension (Yes vs No)	1.554	(0.920,2.624)	0.1			
Diabetes (Yes vs No)	1.086	(0.697,1.691)	0.716			
Clopidogrel (Yes vs No)	0.497	(0.299,0.827)	0.007			
Beta-blockers (Yes vs No)	0.376	(0.244,0.579)	<0.001			
Calcium channel blockers (Yes vs No)	0.579	(0.352,0.952)	0.031	0.376	(0.171,0.826)	0.015
Statins (Yes vs No)	0.152	(0,082,0.282)	<0.001	0.415	(0.182,0.948)	0.037
Low molecular weight heparin (Yes vs No)	0.953	(0.615,1.477)	0.831			
Ticagrelor (Yes vs No)	0.415	(0.207,0.832)	0.013			
Glycoprotein IIb/IIIa receptor antagonist (Yes vs No)	0.603	(0.290,1.255)	0.177			
Angiotensin receptor antagonist (Yes vs No)	0.95	(0.593,1.522)	0.832			
ACEI (Yes vs No)	0.615	(0.398,0.949)	0.028			
Aspirin (Yes vs No)	0.185	(0.118,0.291)	<0.001	0.541	(0.303,0.967)	0.038
PCI (Yes vs No)	0.753	(0.346,1.642)	0.476			

Note: cTnI: cardiac troponin I; CK: creatine kinase; NT-proBNP: N-terminal pro-B-type natriuretic peptide; LDH: lactate dehydrogenase; HDL cholesterol: high-density lipoprotein cholesterol; LDL cholesterol: low-density lipoprotein cholesterol; LVEF: left ventricular ejection fraction; ACEI: angiotensin converting enzyme inhibitor; PCI: percutaneous coronary intervention.

3.4. Predictive power analysis of in-hospital mortality risk prediction model (validation set)

The study found that AUCs of the models for predicting the risk of in-hospital mortality of ACS patients developed with logistic regression, GBDT, random forest, and SVM were 0.884 (95%CI: 0.869, 0.899), 0.918 (95%CI: 0.906, 0.930), 0.913 (95%CI: 0.901, 0.925), and 0.896 (95%CI: 0.884,0.908), respectively (Table 3). The sensitivities were 0.875, 0.938, 0.938, and 0.938 and the specificities were 0.835, 0.812, 0.815 and 0.813, respectively. The predictive performance of GBDT and random forest is better than traditional logistic regression model (Table 3, Fig. 2). We also analyzed the calibration of the model, as shown in the calibration plots (Supplementary Fig. 1), which shows that the model calibration effect was very good, and the calibration power of the model was good.

3.5. Feature importance evaluation of the GBDT and random forest models

Feature importance analysis found that the top 10 variables contributing to the predictive performance of the GBDT model were NT-proBNP (0.242), D-dimer (0.238), Killip (0.184), cTnI (0.064), LDH (0.059), LVEF (0.051), diagnosis (0.042), age (0.035), HDL cholesterol (0.016), CK (0.005). The top 10 variables contributing to the predictive performance of the random forest model were Killip (0.242), D-dimer (0.231), NT-proBNP (0.212), LVEF (0.125), LDH (0.061), cTnI (0.450), diagnosis (0.048), age (0.024), LDL cholesterol (0.004), and HDL cholesterol (0.004), respectively (Table 4, Fig. 3, Fig. 4).

4. Discussion

ACS is one of the main causes of death worldwide [20]. Predicting the mortality risk of ACS patients is helpful for disease management, prolonging the survival time of patients, and improving the quality of life of patients. In our study, the mortality rate was 1.88%, which was lower than that of the Asia-Pacific region; the mortality rate of ACS patients in the Asia-Pacific region during hospitalization was about 5% [21]. The study showed that the models developed using logistic regression, GBDT, random forest, and SVM algorithms can predict the risk of in-hospital mortality of ACS patients. We also found that NT-proBNP, D-Dimer, and Killip were top three variables that contribute the most to the prediction performance of GBDT model and random forest model.

We employed four machine learning algorithms to develop prediction models of mortality during hospitalization in ACS patients and compared their performances. The study found that the prediction performance of GBDT and random forest were superior to the other two types of models. Previous studies have found that the prediction performance of a 3-year mortality risk prediction model for ACS patients

Table 3
Predictive power analysis of each model for predicting in-hospital mortality (validation set).

Models	AUC (95% CI)	P value	Cut-off value	Sensitivity	Specificity
Logistic regression	0.884(0.869,0.899)	<0.001	≥0.435	0.835	0.843
GBDT	0.918(0.906,0.930)	<0.001	≥0.381	0.812	0.908
Random Forest	0.913(0.901,0.925)	<0.001	≥0.486	0.815	0.898
SVM	0.896(0.884,0.908)	<0.001	≥0.227	0.813	0.833

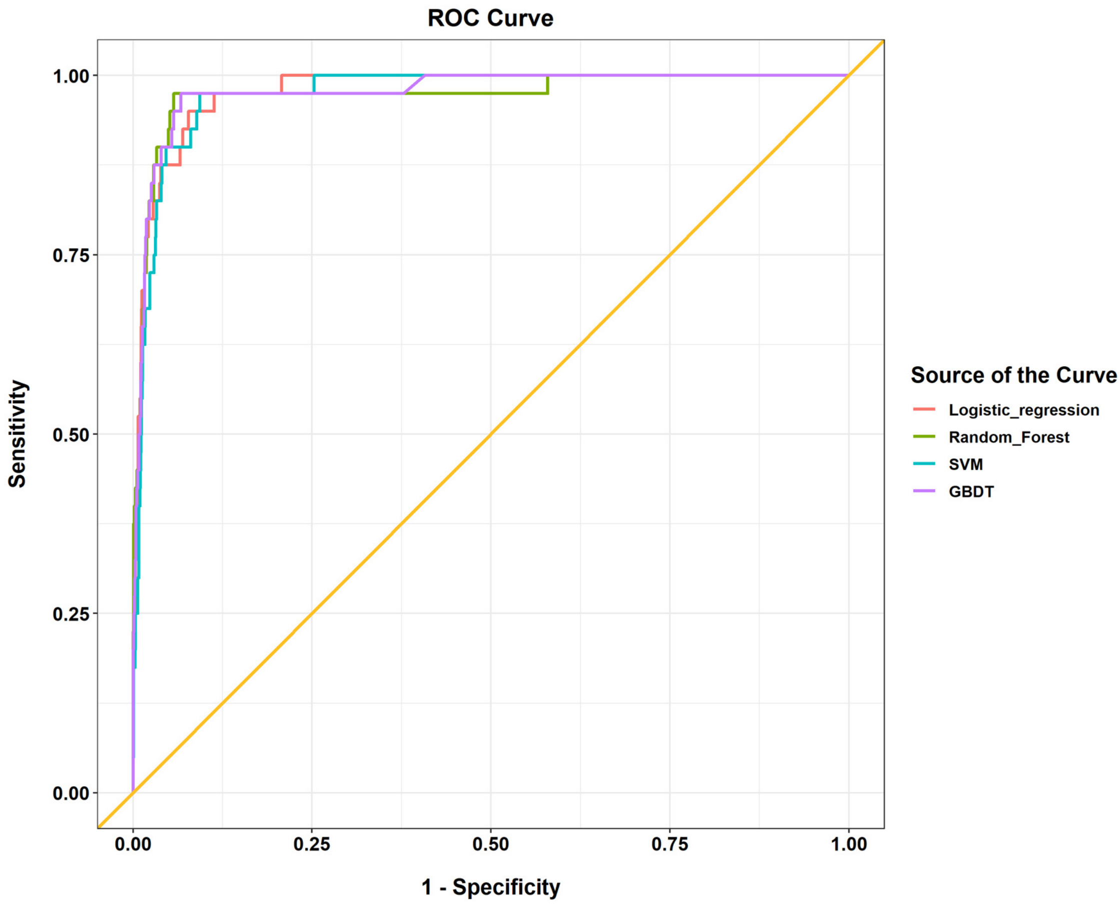

Fig. 2. Receiver operating characteristic (ROC) analysis result of in-hospital mortality risk prediction model.

based on machine learning algorithms was better than the GRACE score (AUC: 0.768 vs. 0.701) [22]. Sherazi et al. found that a mortality risk prediction model based on machine learning algorithms can effectively predict the 1-year death risk of ACS patients [23]. Another study found that a prediction model based on machine learning algorithms can predict the 30-day mortality rate of post-ST-segment elevation myocardial infarction. The authors suggested that machine learning can be used for outcome prediction in complex cardiology settings [24].

The risk of mortality of ACS patients during hospitalization is affected by risk factors. We evaluated the contribution of variables to the predictive effect of the GBDT and random forest risk prediction model. The top three variables that contributed the most to the prediction effect of GBDT and random forest models were the levels of

Table 4
Feature importance evaluation of GBDT model and random forest model.

GBDT		Random Forest	
Feature name	Feature importance (score)	Feature name	Feature importance (score)
NT-proBNP	0.307	Killip	0.242
D-Dimer	0.238	D-Dimer	0.231
Killip	0.184	NT-proBNP	0.212
cTnI	0.064	LVEF	0.125
LDH	0.059	LDH	0.061
LVEF	0.051	Diagnosis	0.048
Diagnosis	0.042	cTnI	0.045
Age	0.035	Age	0.024
HDL cholesterol	0.016	LDL cholesterol	0.004
CK	0.005	HDL cholesterol	0.004

Note: cTnI: cardiac troponin I; CK: creatine kinase; NT-proBNP: N-terminal pro-B-type natriuretic peptide; LDH: lactate dehydrogenase; HDL cholesterol: high-density lipoprotein cholesterol; LDL cholesterol: low-density lipoprotein cholesterol; LVEF: left ventricular ejection fraction.

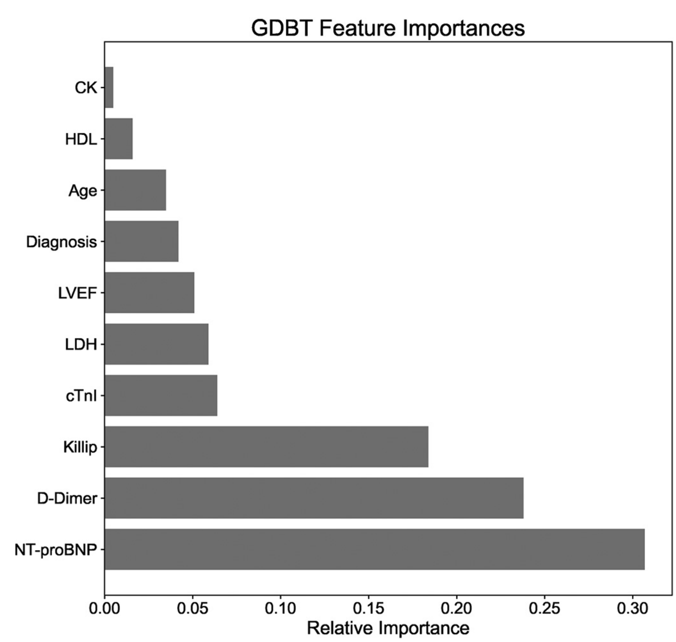

Fig. 3. Feature importance evaluation of the GBDT models.

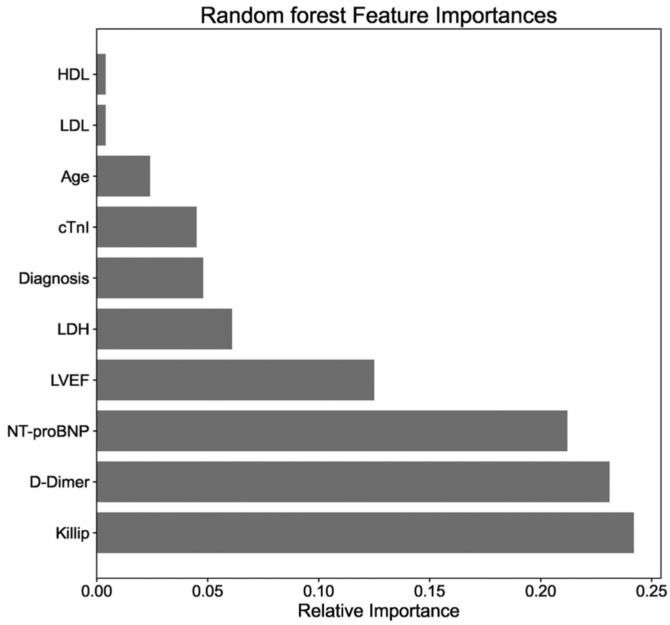

Fig. 4. Feature importance evaluation of the random forest models.

NT-proBNP, D-dimer, and Killip. It was consistent with previous research that has found that among patients with chronic heart failure, patients with increased levels of NT-proBNP had a poorer prognosis [25]. Furthermore, preoperative NT-proBNP can predict the in-hospital mortality and long-term survival of patients undergoing surgery for ACS [26]. Another study found that in non-ST elevation-acute coronary syndrome (NSTE-ACS) patients, NT-proBNP had a good predictive effect on 30-day mortality (AUC $= 0.85$) [27]. ACS patients with higher NT-proBNP levels have a significantly higher 1-year mortality rate than those with lower levels [28], if the patient's NT-proBNP level rises, clinicians must be alert to the occurrence of heart failure, diagnose the patient in time, and take treatment once the disease occurs. The increased D-dimer level was found to be a predictor of a patient's adverse outcome [29]. Several studies have found that increased D-dimer levels were significantly associated with adverse outcomes and increased risk of mortality in ACS patients [30-32]. Killip classification system is an effective risk assessment tool for patients with acute myocardial infarction [33], which is related to the prognosis of patients. The study found that Killip class ≥II, age ≥ 80 years, and brain natriuretic peptide level were significantly associated with a 2-year increase in the risk of death in STEMI patients [34]. Another study confirmed that Killip class can be used as a predictor of short-term and long-term mortality risk in STEMI patients [35]. A prospective cohort study found that older age, Killip class> II, Ejection fraction <40%, and STEMI are independent predictors of 30-day mortality in ACS patients [36]. This is consistent with the results of this study. Therefore, clinicians should closely monitor the D-dimer level and Killip class of ACS patients. Once these two indicators are abnormal, they should take timely interventions to reduce the patient's risk of death.

In addition to the above three important variables, LDH, age, cTnI, and CK, all had important influence on the prediction effect of the model. These results were consistent with previous findings about these factors. Elevated plasma LDH was associated with worse outcomes and increased risk of mortality in patients with several diseases [37-41]. Previous studies have shown that increased plasma LDH level was an independent predictor of the risk of mortality in patients with acute aortic syndromes [42]. Additionally, plasma LDH level was associated with 28-day risk of death in patients with sepsis [43], and in patients with acute decompensated heart failure, plasma LDH may be an independent predictor of 90-day, 180-day, and 365-day all-cause mortality risk [44]. Our study found that LDH was related to the increased risk of mortality in patients with ACS and that it provided a considerable contribution to the predictive effect of the model. Previous studies have found that low levels of HDL and LDL cholesterol have been shown to be important predictors of in-hospital mortality [45-47]. A low early HDL cholesterol level should be regarded as an independent predictor of in-hospital mortality in ACS patients presenting to the cardiac care unit [47]. TRILOGY ACS Trial found that lower baseline HDL cholesterol was significantly associated with increased risk of cardiovascular death and all-cause death in ACS patients [48]. The findings of these studies are consistent with the results of our study. Cardiac troponin I (cTnI) was a validated biomarker for diagnosis and risk stratification of patients with acute coronary syndrome [49]. The increase of high sensitivity-cTnI level in the stable phase after ACS event was an independent predictor of all-cause death and cardiovascular death in the ACS outpatient population [50]. In a stabilized phase of patients with non-ST-segment elevation ACS, cTnI levels exhibited a continuous and slight increase, and levels of cTnI elevated above 0.01 μg/L can predict the long-term mortality of patients [51]. EPICOR registry study found that old age was a risk factor for poor prognosis during the first two years after discharge in ACS patients [52]. As the age of ACS patients increases, the risk of poor prognosis increases, as even elderly patients with good heart function had a higher risk of death [53]. The study found that compared with younger patients, elderly patients with ACS had a higher risk of comorbidities, hospitalization, and 6-month mortality [54]. In our study, we found that in the in-hospital mortality group, more than half of patients were 80 years old or older. Among the surviving patients, most patients are 45–64 years old and 65–79 years old (Supplementary Fig. 1). In addition, a studyfound that increased levels of creatine kinase-MB can predict the risk of hospital death in elderly ACS patients [55].

Our research results suggested that preventing abnormally elevated levels of NT-proBNP, LDH, cTnI, and CK in patients with ACS, while preventing the level of HDL cholesterol in patients from falling below the normal range, may reduce the risk of in-hospital death in patients with ACS. Clinicians should closely monitor the levels of LDH, HDL and NT-proBNP in patients with ACS. Once abnormal levels of these markers are found, relevant measures should be taken in time Intervene to reduce the patient's risk of death. As a marker of myocardial infarction, LDH activity increases to an abnormal level about 10 h after the onset of myocardial infarction, and continues to increase for 6–8 days [56]. Therefore, when the LDH level of a patient is higher than the normal value, clinicians should be alert to the risk of acute myocardial infarction in the patient and take active prevention and treatment. Studies have found that after treatment with gemfibrozil, elevated HDL levels can reduce the incidence of cardiac death and non-fatal myocardial infarction [57]. Another study found that for patients with relatively low baseline HDL serum levels, after receiving a combination of statins and niacin (a drug that raises HDL levels), the risk of adverse cardiovascular events (including death, cardiac Infarction, stroke, and revascularization) was significantly reduced. Compared with the results of single statin therapy in previous studies, the combination therapy in this study reduced the risk of adverse events even more, which showed that increasing the level of HDL gives patients more clinical benefit than just reducing the level of LDL [58]. Therefore, once the patient's HDL level decreases, clinicians should use statin therapy in combination with drugs that increase HDL levels (such as niacin, gemfibrozil, etc.), and adjust the patient's lifestyle, such as Obese patients need to lose weight in order to obtain greater clinical benefits.

This study had several limitations. At first, this was a retrospective study in a single-center which brings bias to the study, and the results need to be verified by a prospective clinical trial. Second, ECG and US findings were not included in this study, which is also one of the limitations. Third, few patients received risk score evaluation and the missing of GRACE scores was more than 40%, thus we could not compare the performance between predictive models and traditional scores like

GRACE scores. Moreover, this study lacked external validation. Finally, although the sample size calculated was not conducted, the required sample size to develop prediction models needed at least 10 events for each predictor parameter [59]. In this study, a total of 25 variables were inclusion, and the sample size should be reached 250–375. The sample size in both training and testing sets were greater than minimum sample size.

5. Conclusion

The predictive models developed using logistic regression, GBDT, random forest, and SVM algorithms can be used to predict the risk of in-hospital death for ACS patients. Based on our findings, we recommend that clinicians focus on monitoring the changes of NT-proBNP, D-dimer, Killip, cTnI, and LDH, as this may improve the clinical outcomes of ACS patients.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ajem.2021.12.070.

Ethics approval and consent to participate

Our study was approved by the Ethical Committee of Fujian Provincial Hospital. The informed consent requirement was waived since the study only involved the use of past clinical data. All methods in our study were carried out in accordance with relevant guidelines and regulations.

Consent for publication

There are no details on individuals reported within the manuscript, consent for publication is not applicable in our study.

Availability of data and materials

All data generated or analyzed during this study are included in this published article [and its supplementary information files].

Competing interests

The authors declare that they have no competing interests.

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Credit author statement

The manuscript has been read and approved by all the authors, all authors of this study met the requirements for authorship of the journal, and that each author believes that the manuscript represents honest work.

The role(s) of all authors:

Jun Ke: responsible for research protocol, data collection and statistics analysis.

Yiwei Chen: responsible for statistics analysis and revised manuscript.

Xiaoping Wang: responsible for retrieval and screening literature, writing research papers.

Zhiyong Wu: responsible for data collection and data review, and final pooling of data.

Qiongyao Zhang: responsible for data collection and data review.

Yangpeng Lian: responsible for supervising the implementation of research and data quality control.

Feng Chen: responsible for the determination of the research direction, the design of the research program.

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